

FOCUSED UPDATES

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Taking a Closer Look at the Role of Inflammation in Cerebrovascular Diseases

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As clinicians, all too often, we are not fully aware of the nuances related to pathophysiology of the diseases we treat. This is both practical and necessary: everyday, treatment decisions have to be made swiftly under time constraints in a chaotic emergency department or on a busy hospital ward to guide patient management, and there is no time to overthink. This holds particularly true in acute stroke: we all know that time is brain,¹ and thus, treatment delays are not acceptable. In this hyper-acute scenario, acute ischemic stroke boils down in our minds to a thrombus in a cerebral artery that needs to be removed (a plumbing problem), aneurysmal subarachnoid hemorrhage is a weakness in the vessel wall that needs to be sealed to prevent further hemorrhage (a leakage problem), and intracranial hemorrhages are simply collections of blood in a closed space exerting mass effect on surrounding structures that, if large enough, need surgical hematoma removal and decompression (a space problem).

Though generally useful, simplifications can sometimes lead to misconceptions and missed therapeutic opportunities. In particular, a purely mechanistic view on acute stroke neglects an important aspect of cerebrovascular disease pathophysiology: the interaction between the brain, particularly the injured brain, and the immune system. Stroke does not occur in isolation from the body—the neurovascular unit interacts constantly, and bilaterally with systemic biology, including the immune system.² Thus, the tissue injury that occurs in a stroke triggers pronounced local and systemic inflammatory processes. Secondary inflammatory processes after stroke follow individual, highly dynamic trajectories and can perpetuate local injury,³ for example through development of vasogenic edema in ischemic stroke and parenchymal

hemorrhage which exacerbates mass effect. This also holds true for the systemic immune response that follows, with immune-mediated secondary effects of stroke leading to poststroke complications such as gut dysbiosis or even sepsis.

This focused update seminar series dives deeper into the role of the immune system and inflammation in acute stroke, highlighting clinical implications and therapeutic opportunities throughout. Vázquez-Reyes et al⁴ set the stage by providing an overview of the immune system of the brain, recent insights and current knowledge gaps. Bosshart et al⁵ build on this further, reviewing how biomarkers allow us to quantify the cerebrovascular immune response. Dias et al⁶ discuss translational barriers and challenges of targeting stroke-related inflammation in acute ischemic stroke. Pensato et al⁷ examine therapeutic opportunities related to thromboinflammation, followed by Frösén and Bhogal,⁸ who explore the role of inflammation in subarachnoid hemorrhage, and the importance of pathogen pattern recognition receptors in this context. Lastly, Khalife et al⁹ outline in their review how recent insights into the role of inflammation in chronic subdural hematoma formation have transformed treatment of this common, and often devastating condition.

After reading this focused update series, it becomes clear how looking further beyond a purely mechanistic view of stroke cannot only help us to better understand stroke cause and mechanisms, but also opens new avenues for novel, targeted treatments of cerebrovascular diseases.

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